BATMAN HAT v1 for Raspberry Pi 4 — schematic generated from hardware/hat-v1/gen/design.py
Design spec: docs/design/hat-v1-spec.md (sections referenced in the part notes)

Blocks (one sheet each):
  1 power_in   battery pads, SMBJ20CA TVS, TPS26631 eFuse + reverse FET, INA226 #1 (0x40), BAT_PRESENT
  2 power_5v   LMR33640 5.17 V / 4 A with hardware UVLO on EN, LM74700 ideal diode to the Pi 5 V pins
  3 power_3v3 TPS62933F FCCM 3.33 V from the Pi 5 V rail, INA226 #2 (0x41), pi filter
  4 softpower LTC2955-2 push-button controller, AUTO-ON jumper, GPIO27 -> KILL, GPIO26 <- INT
  5 pi_header 40-pin header (no ID EEPROM: overlays in config.txt), mounting holes
  6 halow   mPCIe 5.2H socket for Wio-WM6108, bulk capacitors
  7 security   SLB9672 TPM 2.0 on SPI1, ATECC608C-TFLXTLS (0x36), RV-3028-C7 RTC (0x52) + supercap
  8 gnss   u-blox MAX-M10S on UART5 + PPS, active antenna bias, U.FL
  9 debug   Tag-Connect TC2050 (UART0 + HaLow SPI), test points with expected values, power flags

Enable logic: 5 V buck runs when (LTC2955 on) AND (eFuse PGOOD) AND (VSYS > 6.40 V).
              HaLow 3.3 V buck: input = Pi 5 V (HAT buck or Pi USB-C), runs while the Pi 3.3 V is up.
Tier A parts (TPM, ATECC608C, RTC, GNSS) are not assembled by JLCPCB: fitted in Taiwan (spec 5.9).
No test points on TPM / SPI1 nets (spec 5.1).

power_in

1 Power input: TVS, eFuse, reverse protection, system current

File: power_in.kicad_sch

power_5v

2 5 V buck for the Pi + ideal-diode (HAT back-power rule)

File: power_5v.kicad_sch

power_3v3

3 3.3 V for HaLow: source OR-ing, FCCM buck, HaLow current

File: power_3v3.kicad_sch

softpower

4 Soft power: LTC2955-2 push-button / AUTO-ON / KILL

File: softpower.kicad_sch

pi_header

5 Pi 40-pin header, mounting

File: pi_header.kicad_sch

halow

6 mPCIe socket for Wio-WM6108 (pinout unchanged)

File: halow.kicad_sch

security

7 TPM 2.0 SLB9672, ATECC608C, RTC RV-3028

File: security.kicad_sch

gnss

8 GNSS MAX-M10S + active antenna bias

File: gnss.kicad_sch

debug

9 Debug: Tag-Connect, test points, LEDs, power flags

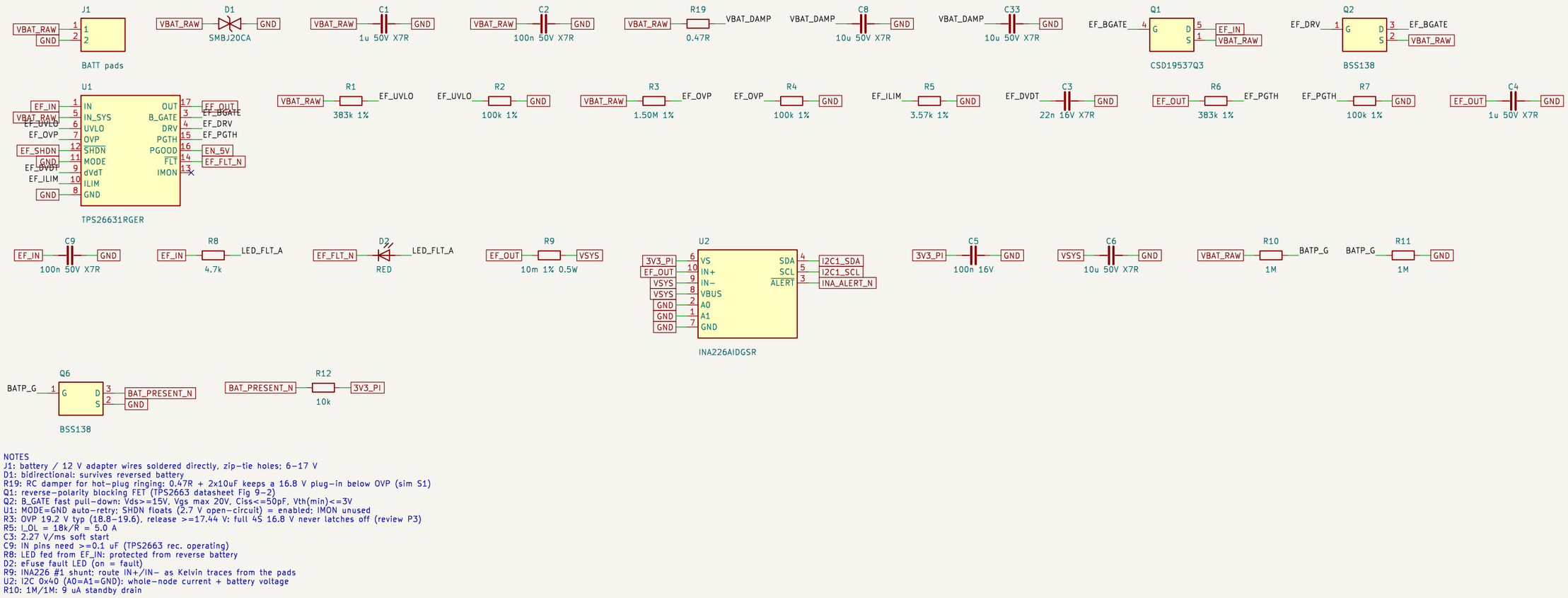
File: debug.kicad_sch

Label-based schematic: same net name = connected
Root: block overview
Batman project — generated by hardware/hat-v1/gen/gen_sch.py, do not edit by hand

Sheet: /
File: batman-hat.kicad_sch

Title: Batman HAT v1 (Pi 4) — replaces Seeed WM1302 HAT

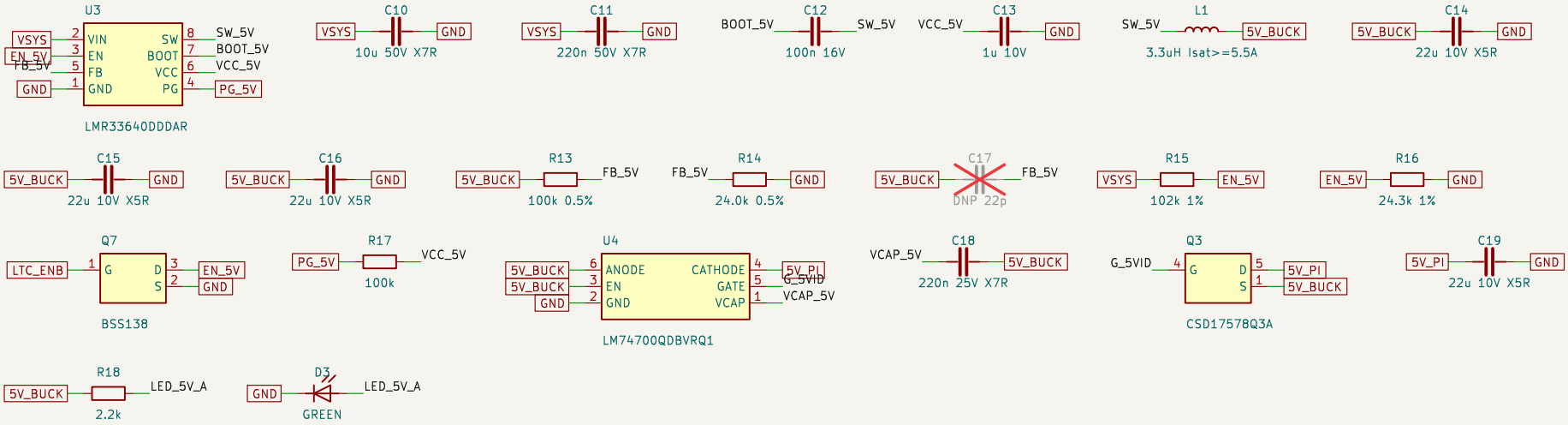
Size: A3	Date: 2026-09-27	**Rev: 0.1-draft**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/10



NOTES
J1: battery / 12 V adapter wires soldered directly, zip-tie holes; 6–17 V
D1: bidirectional: survives reversed battery
R19: RC damper for hot-plug ringing: 0.47R + 2x10uF keeps a 16.8 V plug-in below OVP (sim S1)
Q1: reverse-polarity blocking FET (TPS2663 datasheet Fig 9-2)
Q2: B_GATE fast pull-down: Vds>=15V, Vgs max 20V, Ciss<=50pF, Vth(min)<=3V
U1: MODE=GND auto-retry; SHDN floats (2.7 V open-circuit) = enabled; IMON unused
R3: OVP 19.2 V typ (18.8–19.6), release >=17.44 V: full 4S 16.8 V never latches off (review P3)
R5: I_OL = 18k/R = 5.0 A
C3: 2.27 V/ms soft start
C9: IN pins need >=0.1 uF (TPS2663 rec. operating)
R8: LED fed from EF_IN: protected from reverse battery
D2: eFuse fault LED (on = fault)
R9: INA226 #1 shunt; route IN+/IN− as Kelvin traces from the pads
U2: I2C 0x40 (A0=A1=GND): whole-node current + battery voltage
R10: 1M/1M: 9 uA standby drain

Label-based schematic: same net name = connected
1 Power input: TVS, eFuse, reverse protection, system current
Batman project — generated by hardware/hat-v1/gen/gen_sch.py, do not edit by hand

Sheet: /power_in/		
File: power_in.kicad_sch		
Title: Batman HAT v1 (Pi 4) — replaces Seeed WM1302 HAT		
Size: A3	Date: 2026-09-27	**Rev: 0.1-draft**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 2/10



NOTES
U3: 1 MHz, 4 A
C11: hot-loop cap: closest to VIN/PGND
L1: height <=3.1 mm; alternative: Wurth WE-XHMI 4030
R14: Vout = 1.0 V x (1 + 100/24) = 5.17 V
C17: optional feed-forward (datasheet 9.2.2.8)
R16: EN UVLO: 6.40 V on / 5.88 V off
Q7: LTC2955 EN-bar high (off) -> 5 V buck off
U4: HAT back-power rule: Pi USB-C cannot feed into the HAT 5 V buck
Q3: 30 V, low Rds(on); ideal-diode FET
D3: HAT 5 V buck running

Label-based schematic: same net name = connected
2 5 V buck for the Pi + ideal-diode (HAT back-power rule)
Batman project — generated by hardware/hat-v1/gen/gen_sch.py, do not edit by hand

Sheet: /power_5v/
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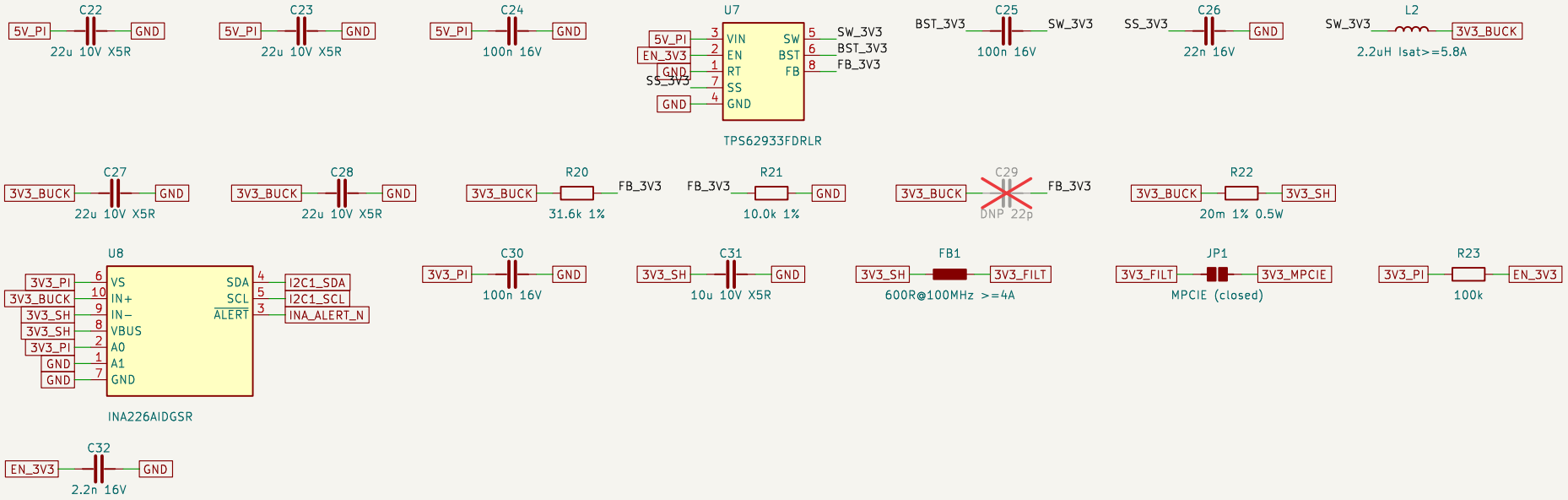
Size: A4	Date: 2026-09-27	**Rev: 0.1-draft**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 3/10



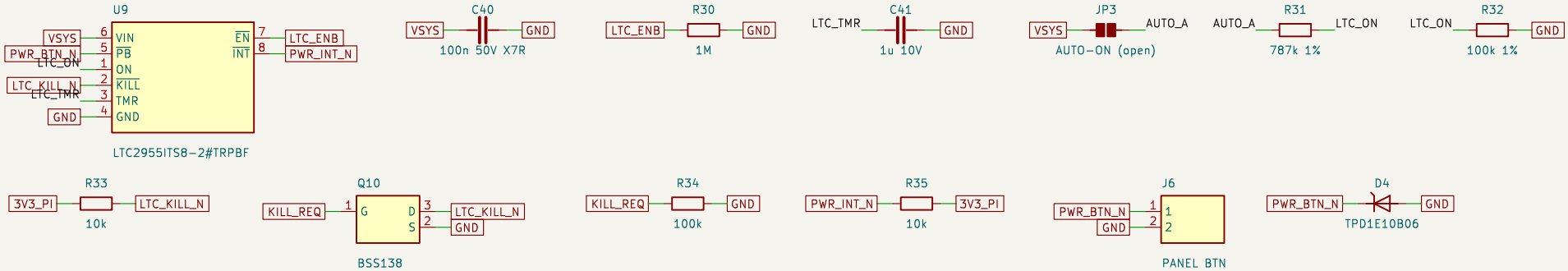
NOTES
C24: hot−loop cap: closest to VIN/GND
U7: FCCM fixed 1.2 MHz (RT=GND); EN abs max 6 V −> open−drain pull−down only
C26: soft start ~3.2 ms
R21: Vout = 0.8 V x (1 + 31.6/10) = 3.33 V
C29: optional feed−forward
R22: INA226 #2 shunt, upstream of every socket bulk cap
U8: I2C 0x41 (A0=VS, A1=GND): HaLow card current
C31: pi−filter input cap
FB1: verify rating at BOM step: >=4 A, <=30 mOhm
JP1: cut to isolate the HaLow card rail
R23: EN pull−up 0.7 uA: Pi off −> EN 0.07 V < 1.1 V off; on −> 3.3 + 0.21 V < 6 V abs max
C32: EN has only a 0.7 uA pull−up: noise immunity

Label−based schematic: same net name = connected
3 3.3 V for HaLow: source OR−ing, FCCM buck, HaLow current
Batman project − generated by hardware/hat−v1/gen/gen_sch.py, do not edit by hand

Sheet: /power_3v3/
File: power_3v3.kicad_sch

Title: Batman HAT v1 (Pi 4) − replaces Seeed WM1302 HAT

Size: A4	Date: 2026−09−27	**Rev: 0.1−draft**
KiCad E.D.A. kicad−cli 7.0.11+dfsg−1build4		Id: 4/10



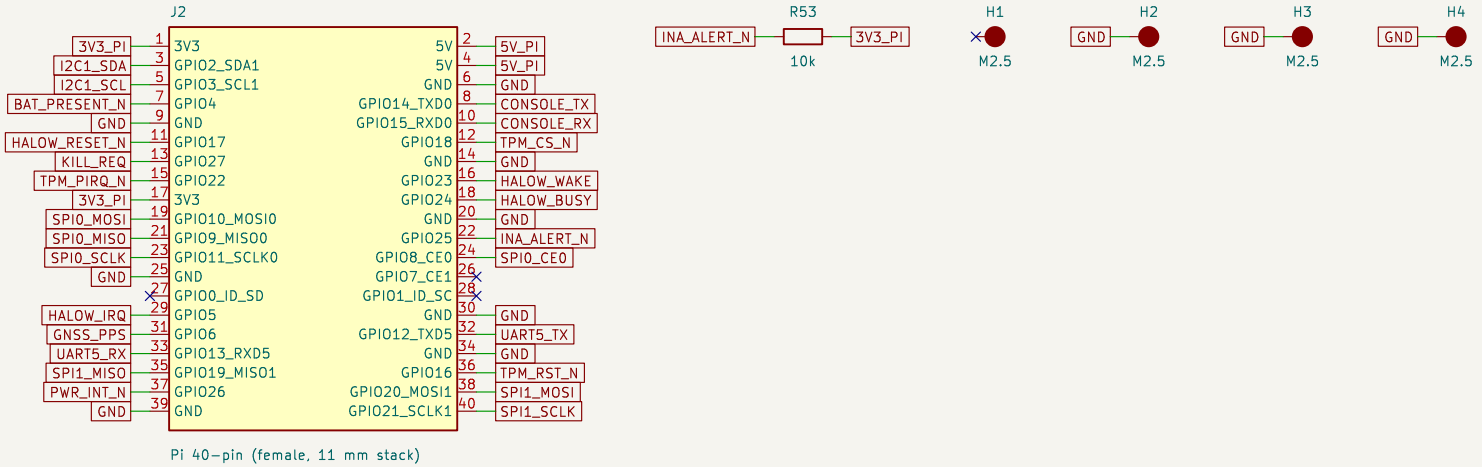
NOTES
U9: -2 = active-low EN-bar (900k to VIN when off); KILL blanking 304-720 ms
R30: gate = 0.53 x VSYS when off (<=9.7 V)
C41: long-press force-off ~5.3 s
JP3: close: power-on automatically when input appears (mast relay / C0 / T0)
R31: AUTO-ON 7.10 V typ / 6.74 V min: above the 5 V buck EN max 6.55 V (review P1)
R33: KILL high as soon as the Pi 3.3 V is up
Q10: GPIO27 high (gpio-poweroff) -> power off
R34: reboot: GPIO27 default pull-down keeps power on
J6: to the IP67 panel push-button. No on-board button (user decision): on the bench plug a button here or short the PWR_BTN_N back-side pad to GND
D4: ESD for the panel button wire

Label-based schematic: same net name = connected
4 Soft power: LTC2955-2 push-button / AUTO-ON / KILL
Batman project - generated by hardware/hat-v1/gen/gen_sch.py, do not edit by hand

Sheet: /softpower/
File: softpower.kicad_sch

Title: Batman HAT v1 (Pi 4) - replaces Seeed WM1302 HAT

Size: A4	Date: 2026-09-27	**Rev: 0.1-draft**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 5/10



NOTES
J2: through-hole on the bottom side; 8.5 mm socket for 11 mm standoffs
H1: Pi HAT mounting hole (NPTH: Wi-Fi antenna keep-out)
H2: Pi HAT mounting hole
H3: Pi HAT mounting hole
H4: Pi HAT mounting hole

Label-based schematic: same net name = connected
5 Pi 40-pin header, mounting
Batman project — generated by hardware/hat-v1/gen/gen_sch.py, do not edit by hand

Sheet: /pi_header/
File: pi_header.kicad_sch

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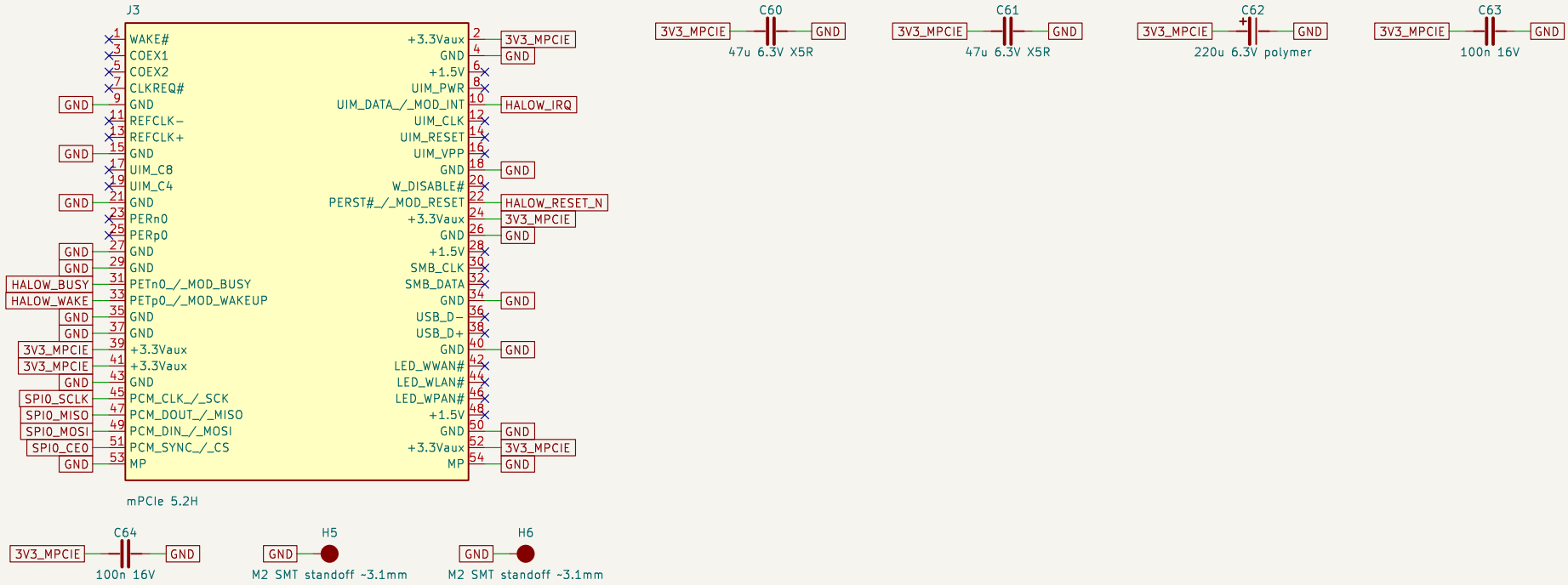
Size: A4	Date: 2026-09-27	**Rev: 0.1-draft**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 6/10



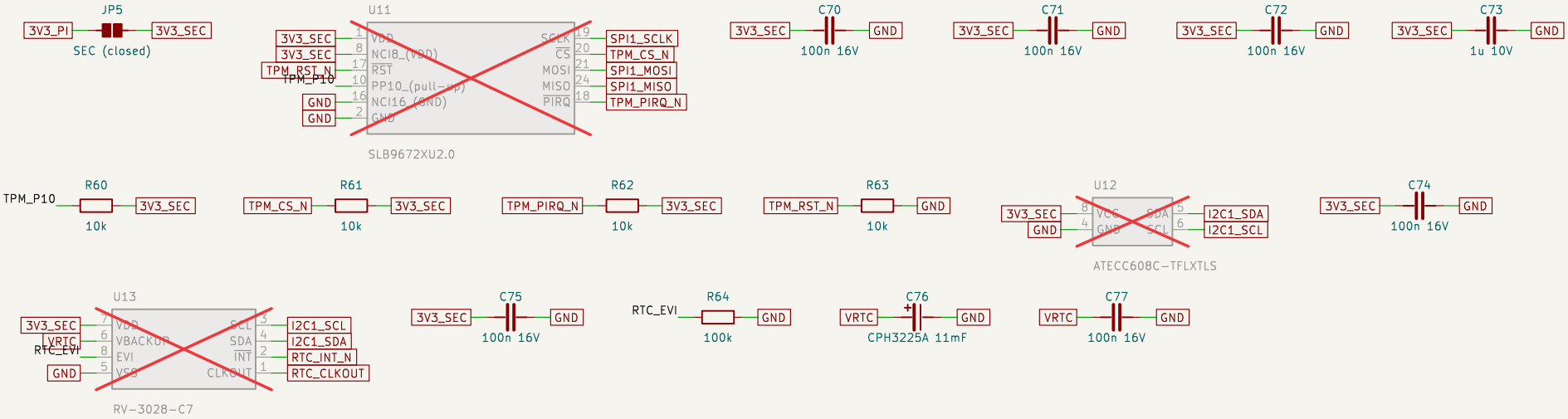
NOTES
J3: pin 8 UIM_PWR not connected on the card −> GPIO18 freed for the TPM
C62: 7343−20: <=1.9 mm tall; absorbs HaLow TX bursts
C63: at pins 2/52
C64: at pins 39/41
H5: card hold−down; height TBD after measuring (spec V14)

Label−based schematic: same net name = connected
6 mPCIe socket for Wio−WM6108 (pinout unchanged)
Batman project − generated by hardware/hat−v1/gen/gen_sch.py, do not edit by hand

Sheet: /halow/
File: halow.kicad_sch

Title: Batman HAT v1 (Pi 4) − replaces Seeed WM1302 HAT

Size: A4	Date: 2026−09−27	**Rev: 0.1−draft**
KiCad E.D.A. kicad−cli 7.0.11+dfsg−1build4		Id: 7/10



NOTES
JP5: cut to isolate TPM / ATECC / RTC supply; I2C/SPI1 still back-power them through ESD diodes
U11: NC 6/29/30 must float; GPIO 3/4/7 have internal pull-ups; NO test points on TPM nets
C70: pin 1
C71: pin 14
C72: pin 22
R60: pin 10 pull-up (TCG compatibility)
R61: datasheet Fig 7: GPIO18 boots pulled down
R63: hold TPM in reset until the gpio-hog releases it
U12: I2C 0x36 (TrustFLEX)
U13: I2C 0x52. App Manual: CLKOUT 32.768 kHz ON by default -> disable in EEPROM (FD=111); backup mode LSM (BSM=11) because VBACKUP ~3.05 V is close to VDD (DSM not recommended, sec 4.2.2)
R64: EVI must not float: tie to VSS through a resistor (App Manual sec 7.1)
C76: RTC backup (user decision 2026-09-27: supercap, GPS re-syncs after ~2 days off). Internal trickle path has a Schottky (0.25 V) -> charges to ~3.05 V, below the 3.3 V rating
C77: VBACKUP decoupling close to the RTC (App Manual sec 7.3)

Label-based schematic: same net name = connected
7 TPM 2.0 SLB9672, ATECC608C, RTC RV-3028
Batman project — generated by hardware/hat-v1/gen/gen_sch.py, do not edit by hand

Sheet: /security/
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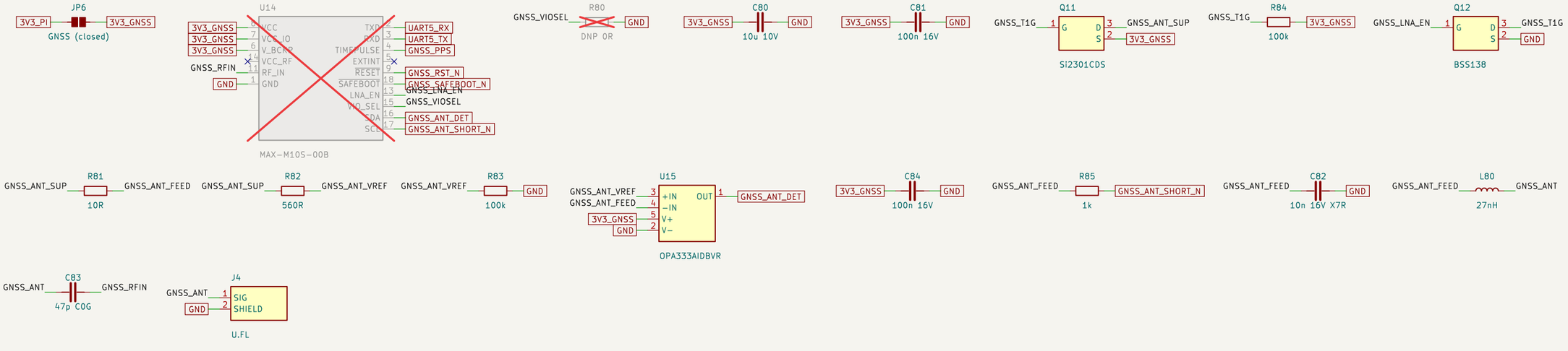
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 8/10



NOTES
JP6: cut to isolate the GNSS supply (UART5_TX idle-high still back-feeds RXD)
U14: UBX-20035208 R08: VIO_SEL open = 3.3 V I/O; V_BCKP 1.65–3.6 V; ext. gain <= 30 dB. SDA/SCL reused as ANT_DETECT/ANT_SHORT_N: CFG-I2C-ENABLED=0 (Integration manual Table 50)
R80: leave NOT fitted: VIO_SEL to GND would select 1.8 V I/O (datasheet Table 10)
Q11: T1: antenna supply switch (reference uses Vishay Si1016X; any small P-FET)
R84: R7: keeps T1 off unless LNA_EN is high
Q12: T2: LNA_EN (ANT_OFF_N) high -> antenna powered
R81: R8: current sense / short limiter 10 ohm 0.25 W (Table 53); -18.5 mV per 1.85 mA
R82: R5 (Table 53)
R83: R6: VREF = 0.9944 x V_ANT -> 'antenna present' above ~1.85 mA
U15: U6 comparator (reference: LT6000/LT6003); zero-drift RRIO, offset <= 10 uV vs an 18.5 mV threshold
C84: U15 decoupling
R85: ANT_SHORT_N taken straight from the R8 low side (reference U7 buffer not needed at equal levels); 1k only limits current into the PIO
C82: C14: bias-T capacitor (Table 52)
L80: L3: bias-T inductor, >500 ohm at L1, >=300 mA (Table 54)
C83: C18: DC block 47 pF 5 % C0G (Table 52)
J4: active patch antenna with SAW pre-filter; 50 ohm CPWG trace

Label-based schematic: same net name = connected
8 GNSS MAX-M10S + active antenna bias
Batman project - generated by hardware/hat-v1/gen/gen_sch.py, do not edit by hand

Sheet: /gnss/		
File: gnss.kicad_sch		
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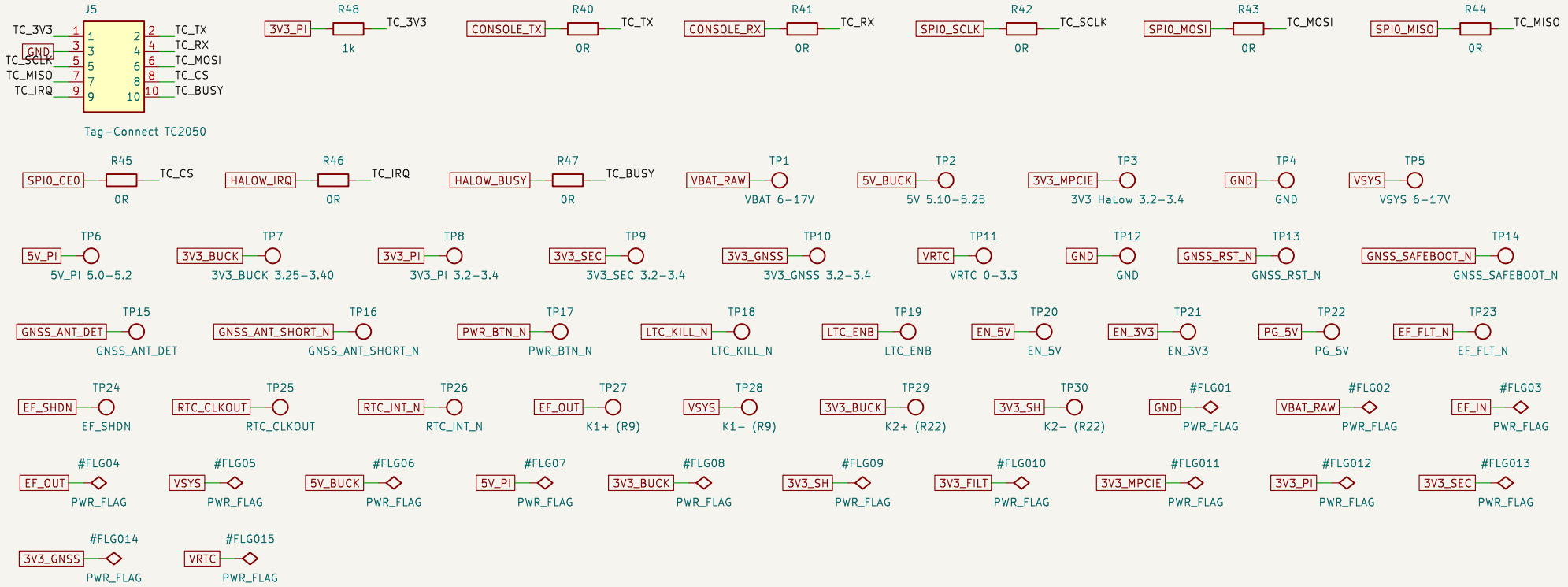
Size: A3	Date: 2026-09-27	**Rev: 0.1-draft**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 9/10



NOTES
J5: console = passwordless root until #137 ships: release gate
R48: sense only: an adapter that drives VCC cannot back-feed the Pi
R40-R47 debug links: fitted on dev boards, NOT fitted on production (HaLow keys cross SPI0)
TP27: Kelvin pads on the shunts: R9 mV x 100 = mA, R22 mV x 50 = mA

Label-based schematic: same net name = connected
9 Debug: Tag-Connect, test points, LEDs, power flags
Batman project — generated by hardware/hat-v1/gen/gen_sch.py, do not edit by hand

Sheet: /debug/
File: debug.kicad_sch

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Size: A4	Date: 2026-09-27	**Rev: 0.1-draft**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 10/10